ALICE L WALTON SCHOOL OF MEDICINE

Cardiac – Compliance, Contractility, Preload, and Afterload

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Adapted from: Contractility, Preload, and Afterload

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0:00 / 28:26 1x

Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Calculate cardiac output using stroke volume and heart rate.
2. Predict compliance of the heart given changes in factors that affect it.
3. Predict contractility of the heart based on changes of factors that control it.
4. Explain preload, what determines it, and relate it to the Frank-Starling law.
5. Predict afterload given changes in factors that affect it.

CASE CONNECTION

You are evaluating NN, a 67-year-old female brought by paramedics to the emergency department (ED). The paramedics report that she has become increasingly short of breath during transport. Her blood pressure is 220/110 mm Hg, her heart rate is 88 beats/minute, and you hear diffuse rales on pulmonary exam. Electrocardiogram is without acute changes. "She's in heart failure," the ED attending says. She turns to you and asks, "What should we do first and why?"

How would you respond to the attending? What therapy would you recommend? Consider these questions as you read this brick, and we'll revisit NN's case at the end.

GO TO CONCLUSION ↓

What Are the Components of Cardiac Output (CO) (LO1)?

Cardiac output (CO) is the primary measure of the heart's pumping efficiency. It represents the amount of blood the heart delivers throughout the body to supply oxygen and re sts its pumping rate and force to match the body's needs during different physiological states like exercise, rest, dehydration, or injury. This brick explains cardiac output, physiological measures (stroke volume and ejection fraction) that affect it, and different laboratory methods to calculate it.

To understand cardiac physiology, it's essential to comprehend how cardiac output (CO) - the volume of blood pumped by the heart per minute - dynamically adjusts to meet the body's metabolic demands. Cardiac output determines both systemic blood pressure and regional blood flow, playing a critical role in tissue perfusion and cardiovascular homeostasis.

Cardiac output (CO) is affected by factors that are intrinsic to the heart (strength of contraction, heart rate) and extrinsic (blood volume and systemic vascular resistance). CO can be expressed in the equation:

$$\text{Cardiac Output (CO)} = \text{Stroke Volume (SV)} \times \text{Heart Rate}$$

Heart rate (60-100 beats/minute) is controlled by the autonomic nervous system—sympathetic stimulation increases rate while parasympathetic decreases it—through modulation of sinoatrial node firing.

Stroke volume is the blood volume ejected per heartbeat (average: 70 mL in males, 20% lower in females). Four main factors influence stroke volume and thus cardiac output.

- **Compliance:** He filled with a volume of blood, like a balloon.
- **Preload:** Passive forces contributing that resist stretching of the heart when filled, reflecting end-diastolic ventricular volume **(EDV)**.
- **Contractility (inotropy):** Strength of heart contraction.
- **Afterload:** The ventricular force generated after ventricular contraction starts, determining blood ejection volume per beat. (Can also be due to stiffened, stenotic pulmonic and aortic valves).

A helpful visual (Figure 1): blood flows from **veins** (determining preload) into the **heart**, whereas contractility, compliance, and heart rate affect the pressures generated and the cardiac output. The ventricles generate pressured high enough to overcome the resistance of the pressure in the arteries, ejecting blood into the circulatory system. These interacting factors—preload, afterload, contractility, and compliance—together regulate cardiac output to meet the body's changing metabolic demands.

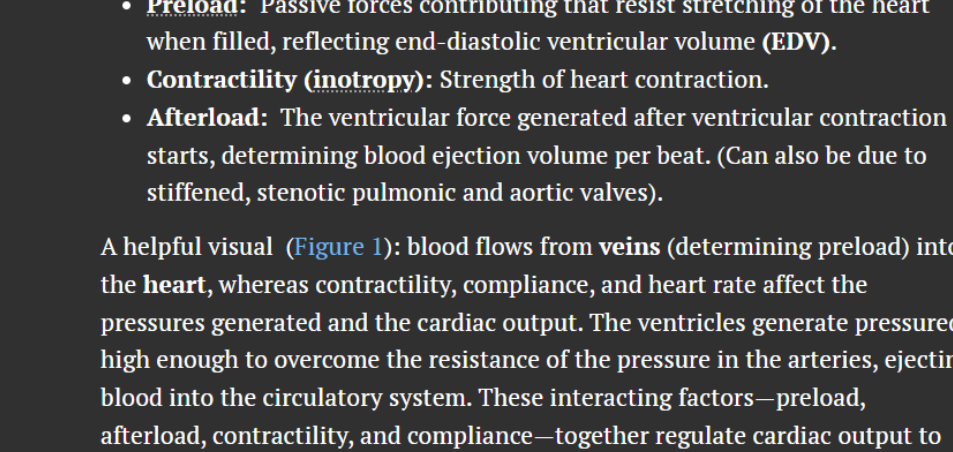
Preload ∝ Volume and Pressures in
Veins
Preload
Heart
HR, Contractility and Compliance
Afterload ∝ Force Pushing blood out
Arteries
Afterload

Figure 1. Visual representation of factors affecting cardiac output

What is Compliance (LO2)?

Cardiac compliance refers to the heart chambers' ability to stretch when filling with blood, similar to a balloon expanding. It's measured as the volume change resulting from a pressure change; where a compliant heart accepts more blood with minimal pressure increase (**end diastolic pressure, EDP**), while a stiff heart resists stretching and fills with less volume. These changes are represented as changes in the compliance curves (also know as end diastolic pressure volume relationship) in Figure 2.

While there is a given level of compliance of the heart, it usually does not become more or less stiff unless there is pathology, such as thickening of ventricles or dilation of the heart, as seen in heart failure. For example, in ventricular hypertrophy, thickened muscle walls become less compliant, requiring higher filling pressures for normal volumes—representing impaired diastolic function.

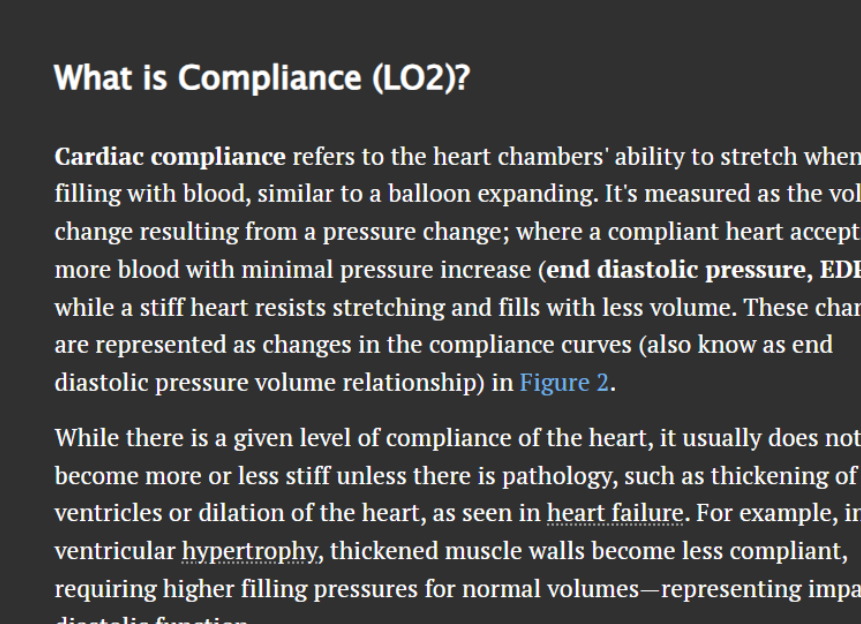

Figure 2. Ventricular compliance alters filling of the heart

What Is Contractility (Inotropy) (LO3)?

Cardiac contractility is the change in ventricular contraction force at a given volume (EDV) or preload. This is seen in Figure 3 where the heart's active force capacity pivots upwards to downwards. Stroke volume and cardiac output are directly proportional to contractility—stronger contractions produce higher output.

Inotropes are substances that modify cardiac muscle contraction force. In healthy individuals, contractility primarily depends on cytoplasmic **calcium** availability. As a positive inotrope, increased calcium levels enhance contractility by promoting the crossbridge cycling, and the actin-myosin interaction that drives contraction.

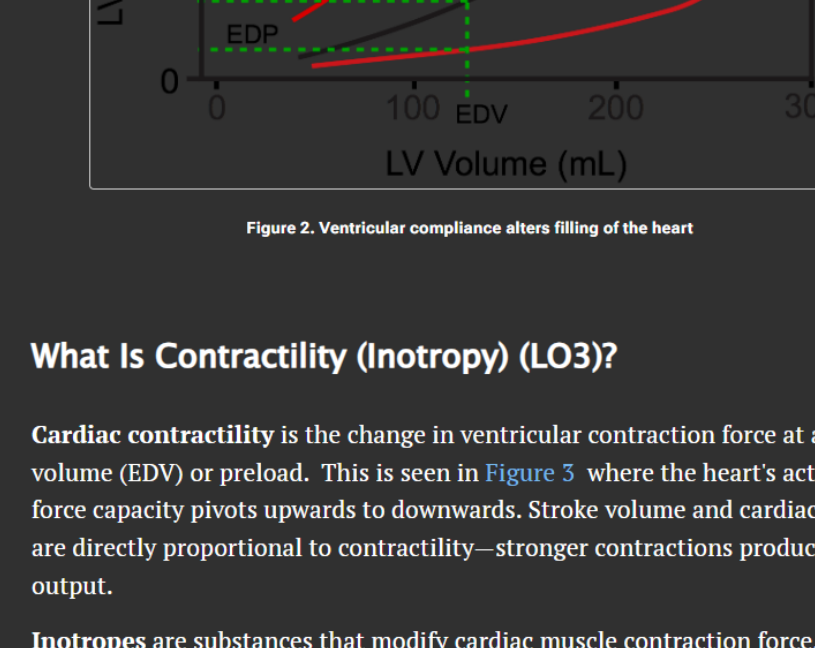

Figure 3: Contractility or force of contraction alters stroke volume

CLINICAL CORRELATION

Digoxin, is an inotrope used to treat heart failure. This drug increases contractility by increasing intracellular calcium.

Estimating Contractility

Contractility isn't directly measured clinically but can be inferred from patient status. A decrease in stroke volume without changes in blood volume or blood pressure suggests declining contractility.

Factors Affecting C

The **autonomic nervous system (ANS)** primarily regulates heart contractility:

- **Sympathetic nervous system** increases atrial and ventricular contractility by activating β1-adrenergic receptors on cardiomyocytes. When norepinephrine binds to these receptors, it triggers signaling that increases intracellular calcium and contractility—an **inotropic** effect.
- **Parasympathetic nervous system** has minimal impact on contractility, exerting only a slight negative inotropic effect limited to the atria, not the ventricles.

What Is Preload (LO4)?

Preload represents factors contributing to passive tension in cardiac muscle before any contraction occurs—essentially the stretch on ventricles at end-diastole. Structurally, it arises from the stretching of connective tissues and degree of overlap of the actin-myosin cross-bridge. Functionally, it depends on venous return: more blood entering ventricles increases stretch and thus preload.

Difference Between Preload and Compliance: Though related, these concepts differ.

- **Preload:** is the actual stretch (volume/pressure stretching heart chambers at end-diastole). This stretch underlies the Frank–Starling mechanism.
- **Compliance:** is heart's "stretchiness"—how easily chambers expand during filling. A highly compliant heart accommodates more volume with minimal pressure increase; a stiff ventricle (low compliance) develops higher pressure for the s g *how much volume* you pour into a container (preload) versus *how flexible* the container's walls are (compliance).

The Length–Tension Relationship

There are three components in the length–tension relationship (Figure 4):

- **Passive Tension**: This is also the preload; the tension develops with the stretching of the muscle.
- **Active Tension**: Tension generated by actin–myosin cross-bridge cycling **after** the muscle is stimulated to contract. At optimal muscle length, more cross-bridges can form, increasing contractile force.
- **Total tension:** The sum of BOTH the passive and active tension.

Like a **rubber band**, pulling the band back (increasing preload) increases tension before it snaps back (active contraction). The more you stretch it, the stronger the snap.

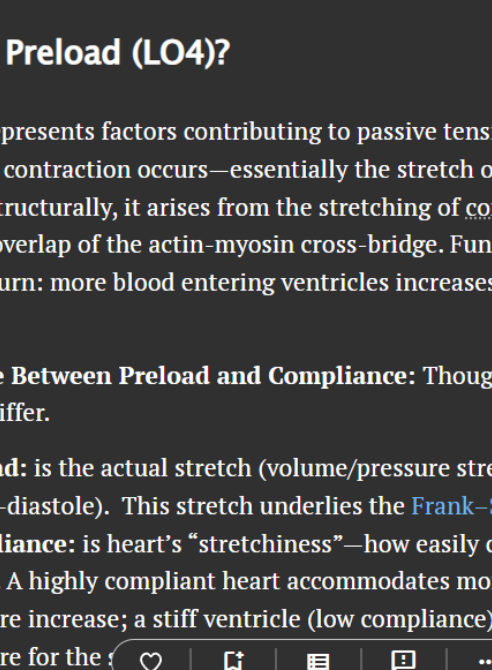

Figure 4. Relationships between Length, Preload (Passive Tension) and Active Tension generated by muscle

What is preload?

Estimating Preload in the Heart

Because the heart is a three-dimensional organ (rather than a single, isolated muscle strip), we typically measure preload using the below:

- **End-Diastolic Volume (EDV)**: Blood volume in the ventricle just before systole (contract
- **End-Diastolic Pressure (EDP)**: Ventricular pressure just before systole.

These two parameters reflect how much the ventricle has stretched during filling. A greater preload increases the force that the cardiomyocyte can generate during systole. High preload **increases the stroke volume**, which, in turn, increases cardiac output.

What two parameters can be used to estimate preload?

The Frank–Starling Law (of the Heart)

The Frank–Starling law states that as preload (ventricular filling) increases, the force of ventricular contraction increases, up to physiological limits (Figure 5). Plotting force at different ventricular volumes produces the Frank–Starling (length–tension) curve. This increase in force occurs because stretching cardiac muscle increases **passive tension** from elastic elements such as titin and places sarcomere ances **active force** through improved cross-bridge formation and increased calcium sensitivity of troponin C. The greater total force allows the ventricle to eject a larger stroke volume. Importantly, inotropic contractility does not change, the heart is simply operating at a different point along the same curve, not on a new shifted curve. This mirrors the classic length–tension relationship seen in isolated muscle fibers (Figure 4), leading to the related conclusion for the heart: **increasing preload increases stroke volume.**

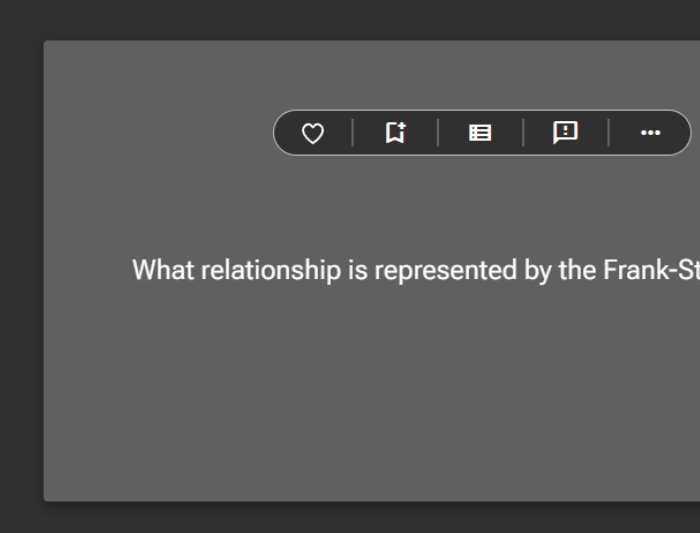

Figure 5. Frank-Starling Law of the Heart: increased preload increases the force of contraction.

What relationship is represented by the Frank-Starling law?

Factors Affecting Preload

Since the venous system fills the heart, anything that increases the venous return of blood to the heart during diastole will increase the end diastolic volume (EDV) and the preload. These include:

- **Blood volume** – the most clinically significant factor. Kidneys regulate this by sodium and water reabsorption under autonomic and hormonal control (particularly the renin-angiotensin-aldosterone system).
- **Central venous pressure** - can increase through greater blood volume (infusion/reabsorption) or sympathetic-triggered contraction of venous smooth muscle venoconstriction). The large veins serve as blood reservoirs (storing ~60% of total blood volume) and, when stimulated they contract and narrow, forcing additional blood to the heart, rapidly stabilizing blood volume fluctuations.
- **Decreased intrathoracic pressure** – enhances the pressure gradient driving venous return (venous pressure – intrathoracic pressure). During deep inspiration, more blood into the thorax. As a result, there is increased venous return, preload, and stroke volume.

CLINICAL CORRELATION

Dehydration or blood loss (e.g., hemorrhage) leads to reduced blood volume and reduced venous return to the heart. Since the heart fills with less blood, the preload is lower, and consequently, stroke volume is lower. Thus, not enough blood is being pumped to the organs, and patients may experience hypotension, dizziness, or reduced organ perfusion.

How will a blood transfusion affect the stroke volume?

What Is Afterlo

Afterload represents factors contributing to ventricular pressure AFTER contraction begins. For the heart, it's the pressure against which ventricles must work to eject blood through the semilunar vavles into the pulmonic and aortic arteries. Higher afterload requires ventricles to generate more pressure for outflow. Stroke volume (and cardiac output) is inversely proportional to afterload.

In Figure 6, points 1-3 representing increasing afterload (y-axis) at the same preload (x-axis). Moving from 1 to 3, the ventricle must generate progressively higher pressure to overcome increasing arterial pressure before opening the aortic valve. This results in ejection beginning at a higher pressure and ending sooner, shortening the ejection phase and reducing stroke volume from loop 1 to 3 (Figure 6). Visually, the top of the pressure-volume loop shifts rightward with increasing afterload (1 to 3), decreasing the width of the loops or the difference between end-diastolic and end-systolic volumes (stroke volume) (Figure 6).

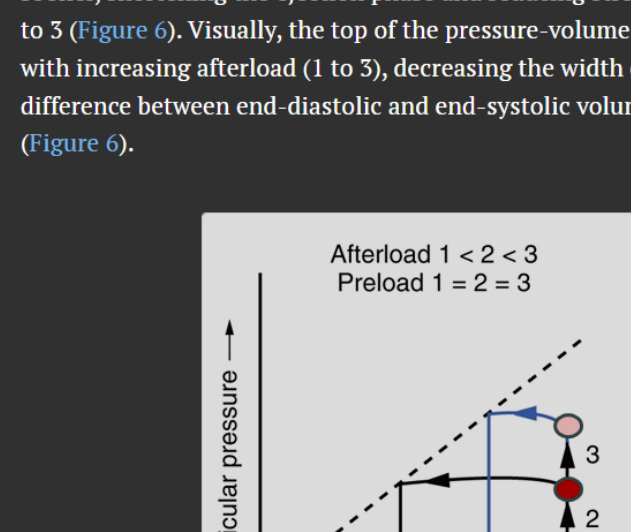

Figure 6. Afterload affects stroke volume

What is afterload?

Estimating Afterload

Under normal conditions, afterload is primarily determined by **systemic vascular resistance** (SVR), often estimated clinically by arterial blood pressure. High afterload is seen with hypertension and also aortic stenosis, and both **reduce stroke volume** (Figure 6).

Why does afterload a erload affects stroke volume because increasing afterload slows myocardial fiber shortening during ventricular contraction. Thus it takes a longer time to generate the pressure to overcome the afterload, and there is less time for blood ejection. Less blood is ejected (smaller stroke volume), leaving more blood in the ventricle at end of systole (contraction).

Factors Affecting Afterload

The main factor that increases afterload is increased resistance, primarily from the contraction of the small arterioles (Figure 1). This resistance is termed systemic vascular resistance (**SVR**) or total peripheral resistance (**TPR**). For example, sympathetic nerve stimulation or circulating hormones (e.g., vasopressin or Ang II) can constrict the vascular smooth muscle that surrounds these small vessels and thus increase SVR. Hypertensive patients often have increased SVR and high afterload. This reduces stroke volume and puts more strain on the heart.

CLINICAL CORRELATION

Aortic stenosis: The valve disease aortic stenosis will also increase afterload. As people age, their aortic valves may become calcified and narrowed. This creates resistance to blood exiting the heart during systole, increasing the afterload. Aortic stenosis will therefore reduce stroke volume and cardiac output, and patients may become dizzy or even pass out (syncope).

How would an increase in afterload affect the stroke volume?

Putting it all together– stroke volume at the center

Figure 7 integrates the four key determinants — preload, compliance, contractility, and afterload — which collectively affect stroke volume.

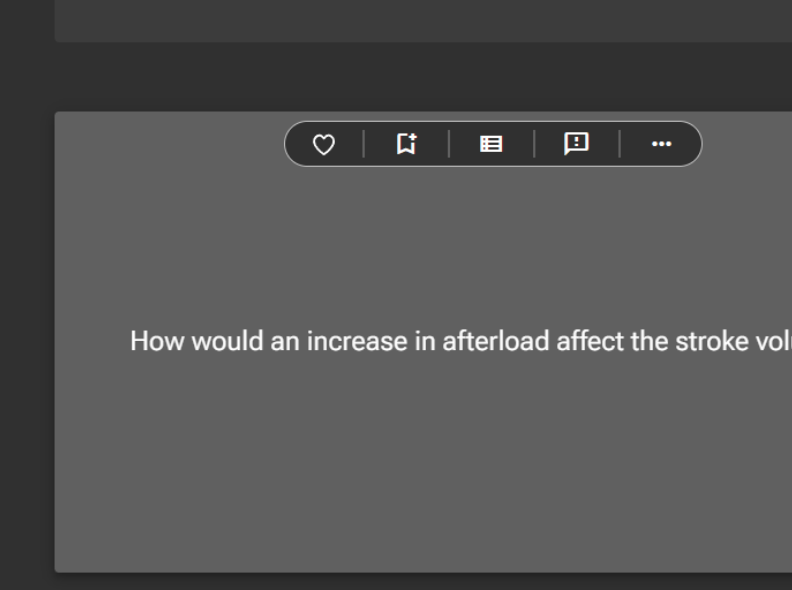
↓ Systemic vascular resistance (SVR)
↓ Afterload
↑ EDV (or preload), ANS input (↑ SNS)
↑ Contractility (Inotropy)
↑ Stroke Volume
↑ Preload
↑ Compliance (to etch)
↑ Central Venous Pressure
↑ Venous return
(including ↓ Intrathoracic Pressure)
↑ Total blood volume

Figure 7. Putting it all together -factors affecting stroke volume

- **Preload**, measured as the volume of blood filling the ventricle during diastole, is influenced by total blood volume and venous return (including effects of decreased intrathoracic pressure). Greater preload results in increased end-diastolic volume.
- **Contractility** represents the heart's intrinsic pumping strength, enhanced by sympathetic nervous system activation. Greater contractility = increased blood ejection from the ventricle.
- **Afterload** is the resistance against which the heart must pump to eject blood, primarily represented by systemic vascular resistance (SVR). Lower afterload facilitates greater stroke volume.
- **Ventricular compliance** reflects the heart's ability to expand during filling. Higher compliance allows the ventricle to accept more blood at lower pressures, enhancing preload without significantly increasing end-diastolic pressure.

Stroke volume results from the balance between ventricular filling factors (preload and compliance), contractile force (inotropy), and ejection resistance (afterload). This integrated perspective explains both normal cardiovascular function and pathological states like heart failure.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to NN, how would you respond to the ED attending? What therapy would you recommend?

You recall that SV is determined by myocardial contractility, preload, and afterload. You tell the attending, "Look at that blood pressure! We need to lower it and reduce afterload," you tell the attending. "If we do that, the heart won't have to work against such high vascular resistance, stroke volume will increase, and her heart failure will improve. I think we should start here." Additionally, the pressure from the systemic circulation is backpropagating to the pulmonary system. The increased pressure forces fluid across the fragile alveolar barriers, causing edema in the lungs.

"Good thinking," she says. After starting NN on an afterload-reducing medication, NN's blood pressure improves, and her shortness of breath resolves. NN is transferred to the medical intensive care unit for continued therapy and does well over the next 24 hours on continued afterload-reducing therapy.

Summary

- Cardiac output (CO) is the volume of blood pumped out of the heart per unit of time; it is calculated as: CO = stroke volume × heart rate.
- Stroke volume (SV) is the volume of blood expelled from the heart during one heartbeat; its three main determinants are contractility, preload, and afterload.
- Heart rate is the number of heartbeats per unit of time.
- Compliance is h ch to accept a given volume of blood, similar to how a balloon expands when filled.
- Contractility is the force generated by the heart to pump the blood at a given preload and afterload; it estimates the intrinsic strength of the heart muscle.
- An increase in sympathetic nerve stimulation results in an increase in intracellular calcium levels, leading to increased contractility and stroke volume.
- An increase in contractility increases cardiac contractile force, SV, and CO; a decrease in contractility does the opposite.
- Preload is the passive force opposing the degree of ventricular stretching at the end of diastole.
- Preload is approximated by the end-diastolic volume (EDV) or end-diastolic pressure (EDP).
- Frank-Starling Law: Stretching of the sarcomeres increases the overlap between actin and myosin; this increases the number of cross-bridges formed and leads to greater contractile force and tension.
- Frank-Starling Law: An increase in preload increases cardiac contractile force, SV, and CO; a decrease in preload does the opposite.
- Afterload is the resistance against which the heart pumps.
- Afterload is approximated by the systemic vascular resistance or systemic blood pressure.
- An increase in afterload lowers the SV and CO; a decrease in afterload does the opposite.

Review Questions

1. A runner is finishing herself to make her fastest time yet. As she runs faster, her heart rate begins to increase to deliver the oxygen her muscles need to continue to propel her forward. How would the contractile force of her heart change and by what mechanism?

   A. ○ Decreases; decreased intracellular Ca^{2+}
   B. ○ Decreases; increased K^+ efflux leads to repolarization
   C. ● Increases; increased intracellular Ca^{2+}
   D. ○ Increases; increased Na^+ influx leads to depolarization
   E. ○ No change in contractility

   Hide Explanation

   The correct answer is C). During exercise, the sympathetic nervous system stimulates the heart to increase cardiac output and supply the exercising muscles with oxygenated blood. This is achieved by catecholamines binding to the β_1-adrenergic receptors on cardiomyocytes and increasing the intracellular Ca^{2+} concentration. Incorrect answers: A). and B). are incorrect because contractility does not decrease in this case. D). and E). are incorrect because neither the K^+ efflux nor the Na^+ influx directly results in any change to contractility.

2. Which of the following parameters describes the resistance exerted on the heart by the pressure present in the aorta?

   A. ● Afterload
   B. ○ Contractility
   C. ○ End-diastolic volume
   D. ○ End-systolic volume
   E. ○ Preload

   Hide Explanation

   The correct answer is A). Afterload is the resistance in the outflow tract that the ventricle has to overcome to pump blood out. Incorrect answers: B). Contractility is the force produced by the heart to expel blood into the outflow channels. C). End-diastolic volume is the total amount of blood in the ventricle at the end of diastole. D). End-systolic volume is the amount of blood left over in the ventricle at the end of systole. E). Preload is the stretch of the myocardium caused by pressure resulting from the volume of blood present in the ventricle.

3. In the context of heart failure, how would an increase in systemic vascular resistance affect the stroke volume?

   A. ○ Increase
   B. ● Decrease
   C. ○ No effect
   D. ○ Increase heart rate
   E. ○ Increase contractility

   Hide Explanation

   The correct answer is (B). Systemic vascular resistance, or afterload, is the resistance the heart must overcome to eject blood. An increase in afterload would make it harder for the heart to pump blood, thus decreasing the stroke volume. Incorrect answers: A). An increase in systemic vascular resistance would not increase the stroke volume. C). It would also not have no effect on the stroke volume. Systemic vascular resistance is not related to D). heart rate or E). contractility.

4. A 70-year-old patient with chronic heart failure presents with worsening fatigue and exercise intolerance. His heart rate is 90 bpm, but due to diminished myocardial function, his stroke volume has decreased by 30% compared to normal. Which of the following best describes the effect on his cardiac output (CO)?

   A. ● Decreases
   B. ○ Increases with HR compensation
   C. ○ Remains unchanged
   D. ○ Decreases only if HR falls
   E. ○ Unaffected by stroke volume

   Hide Explanation

   The correct answer is A). CO = HR × SV; a 30% decrease in stroke volume leads to a proportional decrease in cardiac output, contributing to reduced tissue perfusion. This is clinically relevant in heart failure, where low cardiac output explains symptoms like fatigue and poor exercise tolerance. Incorrect answers: B). Although compensatory mechanisms (e.g., increased heart rate) may occur, the reduced stroke volume directly lowers cardiac output. C). Cardiac output depends on both heart rate and stroke volume. D). The decrease in cardiac output here is directly related to the drop in stroke volume, and not the HR.
   E) .Stroke volume directly influences cardiac output, making this statement inaccurate.

5. A 65-year-old hypertensive patient shows reduced ventricular compliance on echocardiography. Which statement best explains the impact on diastolic filling?

   A. ○ Faster filling rate
   B. ○ Greater contractile force
   C. ● Higher filling pressures
   D. ○ Increased
   E. ○ Reduced end-systolic volume

   Hide Explanation

   The correct answer is C). Reduced compliance means the ventricle is stiffer, leading to higher filling pressures, which can cause diastolic dysfunction and heart failure symptoms such as pulmonary congestion. Incorrect answers: A). Filling rate is not synonymous with compliance; a stiffer heart may fill slower. B). This relates to contractility, not compliance. D). Increased arterial resistance describes afterload, not compliance. E). End-systolic volume is not a measure of compliance. It is the end-diastolic volume (on filling) that would be reduced.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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Objective Confidence: 5/5